Distance Rules Do Not Fit Atolls

Policy brief from DOUBLE JEOPARDY: Do Coastal Rules Reach the People at Risk? Comparing Five Island Jurisdictions · Version 2.1 · Sakshi D. Maske

Key points

- In Lakshadweep and Seychelles, the 20 m and 25 m setbacks cover 7–8% of the buildings standing at or below 1 m, and the Maldives and Fiji have no setback for inhabited land. On all four, the median low building stands 80–136 m from the shoreline. Measured on low land instead of buildings, a 20–25 m line covers about a fifth to a quarter of the ground at or below 1 m on the atolls and Seychelles.
- Widening the setback would not solve this on atolls. Spain's 100 m rule applied to the Maldives would reach 54% of low buildings, but it would also take in 39% of all buildings, because low land on an atoll covers most of the island rather than a strip along the coast.
- The rule itself makes little difference to this. A 20 m line would cover 5–7% of low buildings on every tropical island group, with or without a rule in force, and counting coastal instruments does not show it: Lakshadweep has five of the six types coded in this study and the Maldives two, yet a distance rule fits the low buildings of both equally poorly.
- The Canary Islands are the one case where a setback covers most low buildings. They sit in a narrow beachfront and port strip, and Spain's 100 m line covers 85% of them, though this rests on only 192 buildings.
- Protection can also be reduced. Spain relaxed its coastal law in 2013, and the Maldives widened exemptions from environmental review in 2024–2025.

The problem

A coastal setback sets a minimum distance between buildings and the shoreline, and it is the rule governments use most often to keep new construction away from the sea. It assumes that the land most at risk is the land closest to the water. That holds reasonably well on high islands and continental coasts. On coral atolls the whole island is low, and buildings in the middle of the island are as exposed as those at its edge.

What the study measured

DOUBLE JEOPARDY compared five island groups under five different governments: the Maldives, Lakshadweep (India), Seychelles, Fiji and the Canary Islands (Spain). For about 1.35 million building footprints it measured each building's height on a bare-earth elevation model and its distance to the shoreline, and checked whether buildings at or below 1 m fall inside the setback set by national law. It also coded each government's coastal rules in force in October 2026, along with 23 dated changes to those rules since 1988.

	Maldives	**Lakshadweep**	**Seychelles**	**Fiji**	**Canary Islands**
Statutory setback on inhabited land	none	20 m	25 m	none	20–100 m

	Maldives	Lakshadweep	Seychelles	Fiji	Canary Islands
Coastal instrument types in force (of 6)	2	5	3	3	6
Low buildings within the setback	no setback	6.8%	8.0%	no setback	84.9%
Low buildings within 20 m, same width	5.2%	6.8%	5.6%	6.9%	51.6%
Setback needed to reach 75% of low buildings	151 m	187 m	262 m	262 m	62 m

Recommendations

1. **Set height requirements on atolls and low islands.** Minimum floor or ground levels above a reference water level apply to low buildings wherever they stand on the island. A distance rule is still useful for protecting beaches, but it does little to reduce flood exposure on an atoll.
2. **Test proposed rules against where buildings stand.** With open building footprints and bare-earth elevation models, the share of low buildings a rule would reach can be calculated for any island before the rule is adopted. The calculation is inexpensive and can be repeated each time a rule is revised.
3. **Use bare-earth elevation data.** On forested islands, surface elevation models record the tree canopy as ground level, and can understate the share of people living on low land by a factor of five to eight.
4. **Check instrument counts against design fit.** Comparisons of coastal policy that count laws, plans and agencies can rank a government highly even when its rules are drawn where few of the people at risk live.
5. **Track amendments over time.** Exemptions from impact assessment and longer coastal concessions weaken protection without repealing any rule, and they only become visible when changes are recorded year by year.

Limits

The shoreline used here is the edge of mapped island polygons rather than the legal reference line, so distances may be off by some tens of metres. Elevation has an error of about 0.4–0.5 m. The Lakshadweep setback width and the Maldivian changes of 2024–2025 rest on secondary sources. Many buildings predate the rules, so the analysis shows whether a rule is drawn where low buildings stand, not whether any rule was obeyed or broken.

Full study: Research Paper, Sections 3.9–3.10 and 4.8–4.9 · Code and data: github.com/sakshimaske303-commits/DOUBLE-JEOPARDY · Zenodo DOI: 10.5281/zenodo.23827474